

Deep Learning for Space Time Graph

Assignment 1

Inductive Bias versus Flexibility in Multi-Horizon Forecasting:
From Period-Routed Ridge to Autoformer

<https://github.com/mubeenahmed-ai/AI651-PA1-Forecasting>

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1 Introduction

We study how much a well-chosen inductive bias matters, compared with model flexibility, in multi-horizon time-series forecasting. This was done under two conditions and settings. Task 1 was based on a simulated four-station vibration line whose operating pace changes between runs. On it we compare fixed, sensor-specific and period-routed ridge maps with Attention forecasters, extended by moving-average decomposition and FFT-based delay mixing. A period-routed linear map, which is given the one variable that changes the correct continuation, attains the lowest validation error on both established and held-out stations (MSE 0.028 and 0.030 (m/s²)²). And it is true even when it has 6× fewer parameters than any neural model. Decomposition and delay mixing each help Attention, but their gains are sub-additive. Task 2 forecasts 168 steps of a heavy-tailed real series with an Autoformer. Over 120 multi-seed runs on a chronological split, future-known external variables reduce the mean block RMSE by 18.8 on validation, while past-only co variates give no benefit. A per-horizon diagnosis shows that the reference decoder initialisation ignores the most recent level. A one-parameter learned decay initialisation halves the one-step error and improves validation RMSE on every seed. It is important to note that in both settings, simple models with the right structure match or beat flexible ones.

A forecaster must decide how much structure to assume. Linear maps assume a fixed relation between past and future. Attention [2] on the other hands, lets each context build its own mixing rule. Autoformer [1] assumes that series separate into trend and seasonal parts, and that dependencies are organised by periodic delays. Stronger assumptions make learning easier when they hold and harmful when they do not. Recent work shows that simple linear models often rival Transformer forecasters [3]. This report answers one question in two settings, *how much does the right inductive bias matter compared with a more flexible model?*

Contributions.

1. **Synthetic benchmark (Task 1).** We implement Raw Attention, centred moving-average decomposition, and delay aggregation. We show that period routing, a hand-supplied bias, beats every learned mixer on both populations, and that the two architectural upgrades overlap. The ranking is stable across three model seeds.
2. **Real series (Task 2).** We build an Autoformer from its two defining mechanisms. A 120-run, multi-seed validation study establishes which external variables help, and when their values must be available for them to help.
3. **Decoder initialisation.** We identify a failure of the reference decoder initialisation at short horizons and correct it with a one-parameter learned decay, grounded in the optimal forecast of a short-memory process.

All code, run logs and figures are available at <https://github.com/mubeenahmed-ai/AI651-PA1-Forecasting>.

2 Task 1: Problem Setup

Data. Four accelerometers (20 Hz, 15,360 samples each) record

$$y_i(t) = b_i(t) + s_i(t) + \epsilon_i(t),$$

i.e. a slow drift, a vibration, and noise with $\sigma = 0.09$.

- **Products.** Products A, B, C alternate every 192 samples. Each run has one vibration period, drawn from its product’s range (14–18, 21–27, 28–36 samples) and shared by all stations.
- **Drift.** The drift has period 240. Over one context it spans up to 9.1 m/s², against 3.5 for the vibration.

- **Populations.** Stations 1–3 are *established* (EST). Station 4 is *held out* (HO): it is never used for fitting.

Protocol.

- Context $L = 96$, horizon $H = 48$. No window crosses a changeover.
- Targets lie in train $[0, 9600)$, validation $[9600, 12480)$ and test $[12480, 15360)$.
- Each context is anchored at its last value a and divided by one training scale c ; forecasts are $\hat{y} = a + c\hat{z}$.
- All results come from one fresh PA1_PRESET=full run (data and model seed 0). Every harness check passed.
- Unless stated otherwise, errors are *validation* errors in physical units.

3 Task 1: Methods

Linear baselines. *Shared ridge* fits one $\widehat{W}$ to all EST windows. *Sensor-specific ridge* fits one per established station. *Period-routed ridge* estimates the context’s period, snaps it to one of 13 fixed 1.5-sample pace bands, and uses that band’s $\widehat{W}$.

Raw Attention A circular 3-tap convolution embeds the context to width $d = 64$. A learned position table is added, followed by two pre-norm encoder blocks (Attention and feed-forward, with residual connections), a LayerNorm, flattening, and a linear head to H outputs: 368,368 parameters.

Series decomposition (implemented). For odd width $m = 2q + 1$, the trend is

$$T_t = \frac{1}{m} \sum_{u=-q}^q x_{t+u},$$

computed after replicating the first and last samples q times. We return $(S, T) = (x - T, T)$, which makes $S + T = x$ exact. The decomposed forecaster sends S to the encoder and T to a linear trend head, adds the two forecasts, and re-decomposes after every residual sub-layer.

Delay mixer (aggregation implemented). Scores $R(\tau)$ are computed on projected q, k (supplied). Delays are restricted to $\{8, \dots, 48\}$, and the top $K = \lceil 2 \ln 96 \rceil = 10$ are kept with weights $\alpha_j = \text{softmax}_j(R(\tau_j)/\vartheta)$ at a learned temperature ϑ . We implement

$$z_t = \sum_{j=1}^K \alpha_j v_{(t-\tau_j) \bmod L} \tag{1}$$

as a per-example gather along time. Delays and weights are shared across heads and features, and the operation is differentiable in both v and α . On the worked example ($v = [10, 20, 30, 40]$, delays $[1, 3]$, weights $[\frac{3}{4}, \frac{1}{4}]$) it returns $[35, 15, 25, 25]$, which confirms the shift direction. A single network toggles two switches: representation (raw or decomposed) and mixer (Attention or delay). This gives Raw Attention, Attention + decomposition, Raw delay mixer, and Autoformer-inspired.

4 Task 1: Results

4.1 Context-dependent Forecasting (Responses 1A, 1B)

Response 1A (expected ordering). We derived an expected ordering from each model’s assumptions as the handout describes them:

Table 1: Validation errors (Outputs 1.1 and 4.1). RMSE in m/s^2 , MSE in $(\text{m/s}^2)^2$. EST = Stations 1–3; HO = held-out Station 4.

Model	EST					HO		Params	Fit (s)
	RMSE	MSE	A	B	C	RMSE	MSE		
Shared ridge	0.767	0.588	0.805	0.854	0.621	0.876	0.767	4,608	<0.01
Sensor-specific ridge	0.768	0.590	0.804	0.852	0.630	n/a	n/a	13,824	<0.01
Period-routed ridge	0.166	0.028	0.181	0.164	0.152	0.173	0.030	59,904	<0.01
Raw Attention	0.424	0.180	0.387	0.463	0.419	0.621	0.385	368,368	707
Attention + decomposition	0.299	0.089	0.268	0.341	0.282	0.444	0.197	373,024	745
Raw delay mixer	0.217	0.047	0.208	0.207	0.234	0.340	0.116	368,370	736
Autoformer-inspired	0.199	0.039	0.223	0.189	0.182	0.240	0.058	373,026	758

- EST: period-routed < Raw Attention < shared $\approx$ sensor-specific. Routing supplies the variable that changes the rule; Attention adapts but must learn how; one fixed map must compromise across paces; station identity carries no pace information.
- HO: the same order without sensor-specific ridge, with some degradation for every model and most for the neural one.

Response 1A (Measured, Table 1).

- EST: period-routed (0.166) < Raw Attention (0.424) < shared (0.767) $\approx$ sensor-specific (0.768).
- HO: $0.173 < 0.621 < 0.876$.
- **Magnitudes.** Routing cuts shared-ridge RMSE by 78% and beats Raw Attention by $2.6\times$ on EST. From EST to HO, Raw Attention degrades by 46%, against only 4% for routed ridge.
- **Products.** Shared ridge is worst on Product B (0.854) and best on C (0.621). Routed ridge is flat (0.152–0.181).

Response 1A (Revision). The ordering was confirmed, but two magnitudes were revised upward. The first is routing’s margin over Attention. The second is Attention’s held-out loss. That loss exceeds pure amplitude scaling: anchored linear maps are scale-equivariant, and shared ridge follows amplitude exactly (0.731 on Station 1 at amplitude 1.75, rising to 0.876 on Station 4 at 2.15, against 0.898 predicted). Scaling would put Raw Attention near 0.45 on HO, yet it is 0.621, which indicates overfitting to three stations.

Response 1A (Explanation).

- One map must average incompatible phase advances.
- Identity is uninformative, because each station runs every pace.
- Routing conditions on pace alone, so it transfers to HO.
- Attention adapts per context, but must learn pace and drift from limited data.

Response 1B. The correct continuation advances the phase by $2\pi h/P$, so $P \approx 16$ and $P \approx 30$ require different linear operators. The single least-squares $\widehat{W}$ of Eq. (8) in the handout averages them, and an average of phase-shifted sinusoids is a *damped, phase-smeared* sinusoid (Fig. 1, top).

- **Product A.** Shared ridge’s forecast loses amplitude and drifts out of phase: window RMSE 0.779, against 0.232 for routed ridge.
- **Product C.** The phase error is smaller: 0.448 vs 0.150.
- **Sensor-specific ridge** shows the same shape (0.787, 0.436).

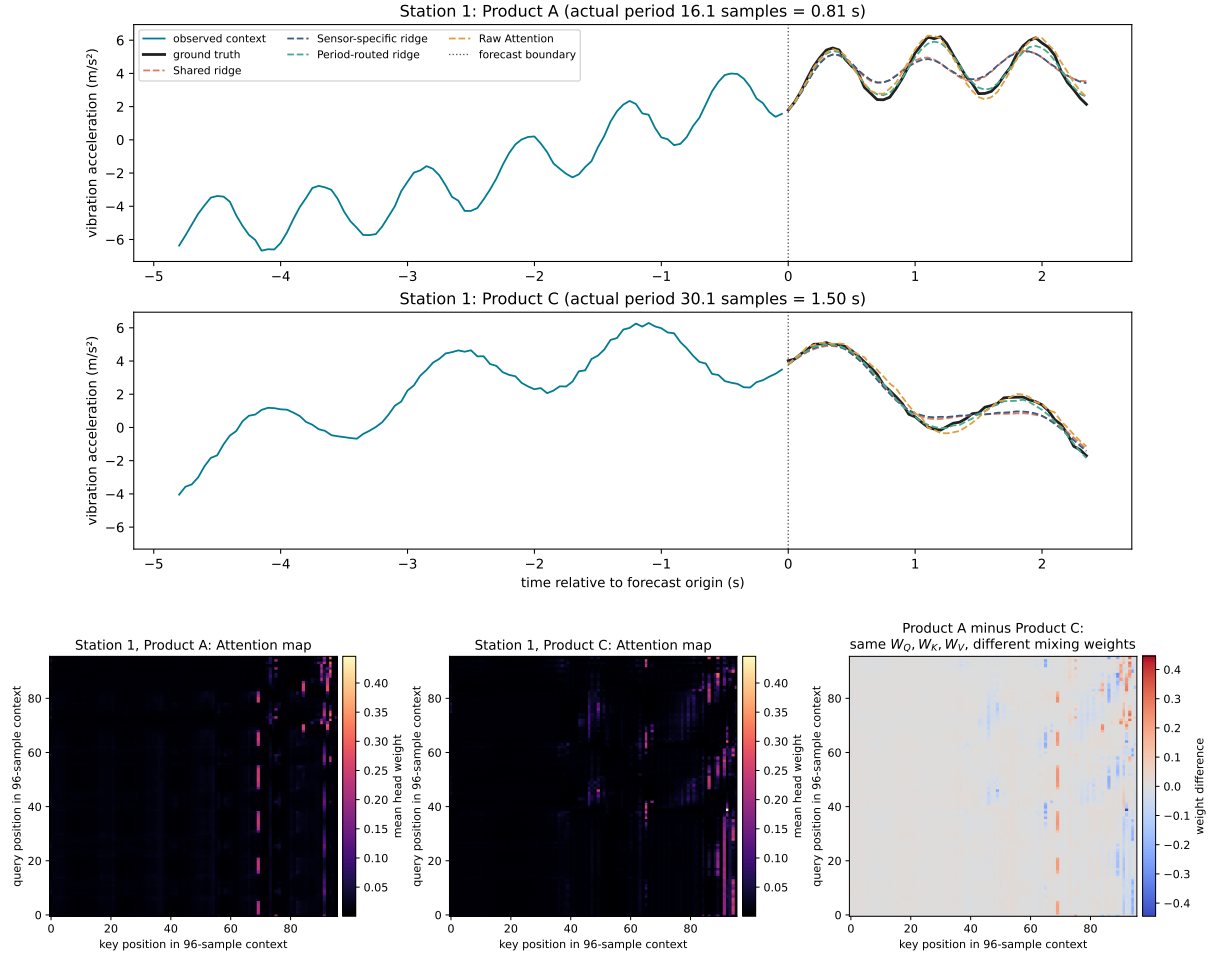


Figure 1: Top: Output 1.2, Station 1 under Product A ($P = 16.1$) and Product C ($P = 30.1$). Bottom: Output 1.3, first-layer Attention maps for the same station under both products, and their difference.

In contrast, Attention keeps W_Q, W_K, W_V fixed but recomputes $QK^\top$ from each window. The two contexts therefore yield different mixing operators without refitting: mean $|\Delta| = 0.011$, comparable to the uniform weight $1/96$, and max 0.445 (Fig. 1, bottom). This shows only that the mixing depends on the input, not that Attention has recovered the period.

4.2 Separating Slow Structure

Selected width. We selected $m = 41$.

- **Oracle recovery** (training, EST; Output 2.2) rises from 0.687 on the raw context to 0.980, 0.989, 0.993 and 0.998 for $m = 17, 25, 33, 41$.
- **Visually** (Output 2.1, Fig. 2), narrow widths let the trend follow the vibration: at $m = 17$ the residual std is 0.58, against 1.57 at $m = 41$. After decomposition, the periodogram peaks sharply at the true $P = 32$.
- **Empirically** (Output 2.3), Attention + decomposition halves Raw Attention’s validation MSE: 0.180→0.089 on EST, 0.385→0.197 on HO, and lower on every product.

Why the bias fits and Failure Mode The drift (period 240) and the vibration (14–36) are separated in frequency. As, $m = 41$ keeps 95% of the drift in T while leaking on average 14% of the vibration. At $m = 17$ the leak is 37% on average and up to 67% at $P = 36$, matching Fig. 2. The oracle measures directly whether a width exposes the repetition, without training one network per width, and it uses training windows only. It saturates, however, and $m = 41$ is the largest candidate, so the choice is “best of four”, not a proven optimum. An abrupt level shift is smeared over m samples and leaks into S . STL [5] would instead assume a locally smooth but curved trend.

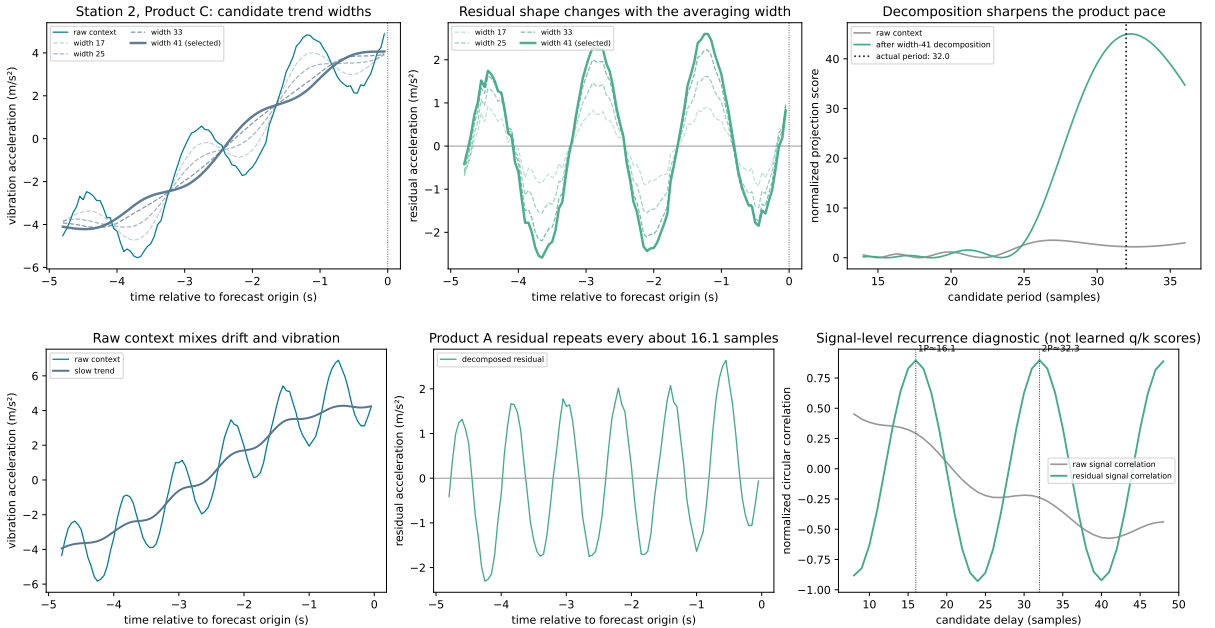


Figure 2: Top: Output 2.1, candidate trend widths on the validation window with the *largest* raw period error (4.98 samples vs a median of 0.53), i.e. a hardest case. Bottom: Output 3.2, signal-level recurrence diagnostic (not learned q/k scores) for a Product A run with $P = 16.13$.

4.3 Mixing by Temporal Delay

Implementation. Eq. (1) passes the worked example and the gradient and axis checks. Output 3.1 confirms that the supplied FFT scores equal the direct circular correlation, $[4, -4, 4, -4]$.

Strongest lags. The residual correlation peaks at delays 16 (0.896) and 32 (0.897), the nearest integers to $P = 16.13$ and $2P$, with troughs near $P/2$ (Fig. 2, bottom). This is the $\cos(2\pi\tau/P)$ profile expected for a sinusoid. The near-equal peaks reflect $96/P \approx 5.95$ cycles fitting the circular ring.

Why Decomposition helps and Failure Mode A ramp resembles itself at every shift, so the raw correlation decays smoothly with no peak at P . Removing the drift isolates the recurring displacement, supporting a mixer that scores L delays rather than L^2 position pairs. An isolated transient, such as an impact, has no recurring displacement. Delay mixing would then forecast a spurious echo one pseudo-period later. A period drifting within the window fails similarly.

4.4 Comparing Designs and Deployment

(a) Prediction:

Decomposition should help most on contexts where the slow component is prominent, because it removes the ramp before the mixer and hands it to a linear trend head. Delay mixing only changes how a still drift-dominated sequence is mixed, and a large ramp makes every delay look alike.

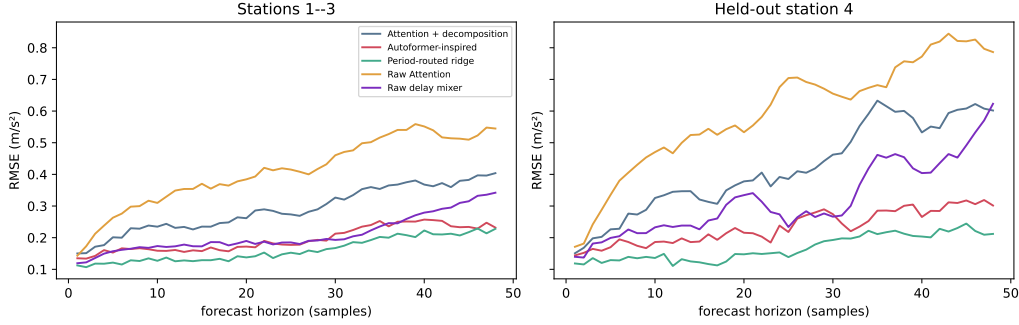


Figure 3: Output 4.2: validation RMSE along the 48-step horizon, including period-routed ridge.

(b) Errors. Validation RMSE and MSE for the four neural models and the linear baselines on EST are given in Table 1.

Interpretation. Measured as validation MSE gain over Raw Attention, decomposition gives -0.091 on EST, delay mixing -0.133 and both together -0.141 , against a sum of -0.224 . On HO the gains are -0.188 , -0.269 and -0.327 , against a sum of -0.457 . Both switches help, but their gains are sub-additive because both target the drift. Delay mixing is the stronger single switch. On EST, RMSE grows along the horizon (Fig. 3): from 0.14 to 0.55 for Raw Attention, 0.14 to 0.23 for the Autoformer-inspired model, and 0.11 to 0.23 for period-routed ridge. Prediction (a) is not supported. On the quarter of EST validation windows with the largest slow-component range, delay mixing gains -0.083 against -0.018 for decomposition. These windows are near-linear ramps, which anchored models already extrapolate. Decomposition’s gain comes instead from curved-drift windows (-0.115). **(c) Deployment.** Period-routed ridge for both populations, chosen from the validation-only summary before Output 4.3.

- **Established stations.** Lowest validation MSE (0.028 vs 0.039 for Autoformer-inspired), $6\times$ fewer parameters, and a closed-form fit (<0.01 s vs 758 s).
- **Held-out Station 4.** Again lowest (0.030 vs 0.058). It routes by estimated pace, not identity, so it needs no Station-4 data.
- **Test (Output 4.3).** RMSE 0.161 (EST) and 0.183 (HO), matching validation. Model seeds 1–2 give the same ranking.

4.5 Robustness

Model seeds. Retraining the four neural models with seeds 1 and 2 preserves the ranking on both populations. Over seeds 0–2, the mean $\pm$ std EST validation MSE is 0.192 ± 0.012 for Raw Attention, 0.071 ± 0.017 for Attention + decomposition, 0.042 ± 0.006 for the raw delay mixer, and 0.036 ± 0.004 for the Autoformer-inspired model. Period-routed ridge reaches 0.028; it is a closed-form fit, so it has no seed spread. Sub-additivity also persists. Decomposition and delay mixing gain -0.121 and -0.150 separately, but only -0.156 combined, well short of their sum of -0.271 .

5 Task 2: Problem and Data

The task is to forecast $H = 168$ steps (time_idx 43,657–43,824) of a non-negative series with 43,656 released observations.

- **External file.** Ten variables A–J, measured on the same clock, cover the horizon as well. Using their horizon values is permitted.
- **Constraints.** The model must be an Autoformer with progressive series decomposition and Auto-Correlation. Claims require multiple seeds.
- **Score.** Ranking uses $\text{RMSE} + \alpha P + \beta E$.

Exploratory analysis (training split only).

- **Periodicity.** Without calendar information, the periodogram reveals a 24-step cycle in the target and all continuous covariates, and a $\approx 8,766$ -step cycle, the dominant peak of features A–C (Fig. 4). These are the daily and annual cycles of hourly data.
- **Distribution.** The target is heavy-tailed: median 73, mean 98, maximum 994, skew 1.8.
- **Memory.** Autocorrelation is 0.97 at lag 1, 0.40 at lag 24 and ≈ 0 at lag 168. Over a week, a constant training mean therefore beats persistence (validation block RMSE 85.9 vs 105.6).
- **Covariates.** Indicators G–J are one-hot. The strongest covariate links are dew point A (0.67 correlation of 24-step changes), pressure C (-0.42) and cumulative wind D (-0.30).

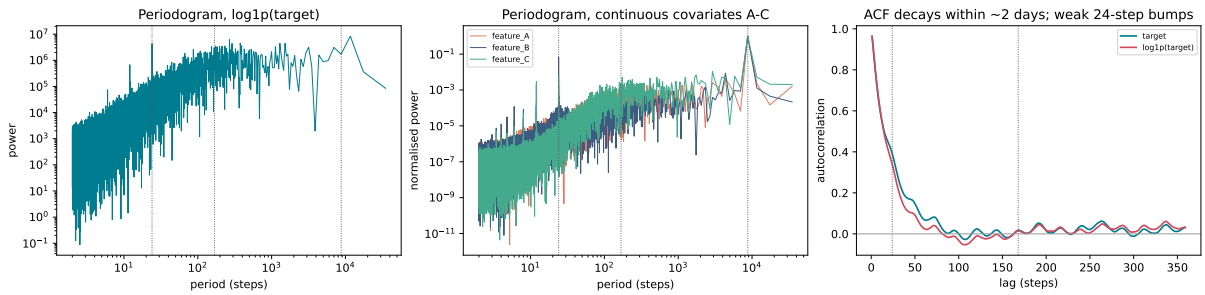


Figure 4: Task 2, training split: periodogram of $\log(1 + y)$ and of covariates A–C, and autocorrelation. Dotted lines mark 24, 168 and 8,766 steps.

6 Task 2: Method

Preprocessing.

- The target is z-scored on the raw scale with training statistics. Forecasts are clipped at 0.
- Continuous covariates are z-scored, after $\log(1 + \cdot)$ for the heavy-tailed D–F. Indicators stay binary.

- Phase features $\sin / \cos(2\pi t/24)$ and $\sin / \cos(2\pi t/8766)$ are computed from the index, so they are known over the horizon.

Covariate regimes. These define the ablation:

- *future*: the measured horizon values;
- *past*: known only to the origin, held at their last value over the horizon;
- *none*.

Covariates enter as extra *value channels*: in the encoder with the target, and in the decoder with their known future values.

Autoformer. Our implementation follows [1] and its reference code (github.com/thuml/Autoformer, MIT licence)..

- **Series decomposition.** Edge-replicated moving averages of width 25 decompose the input, and follow every Auto-Correlation and feed-forward sub-layer. The encoder keeps the seasonal part. The decoder accumulates its three extracted trends per layer into the trend stream.
- **Auto-Correlation.** Delay scores are computed by FFT and averaged over heads and features. The top $k = \lfloor \ln L \rfloor$ delays are softmaxed and aggregated by circular roll, $v_{t-\tau}$. There is no point-wise attention.
- **Configuration.** Input 168, label 48, horizon 168; $d_{\text{model}} = 16$, $d_{ff} = 32$, 4 heads, dropout 0.1.
- **Deviations from [1].**
 1. Delays are selected per sample during training as well as inference.
 2. Covariates enter as value channels.
 3. The decoder’s horizon trend starts from a learned blend (below).

Learned Decoder Initialisation. The reference decoder starts the horizon trend at the encoder-window mean. For a short-memory process, however, the optimal h -step forecast relaxes from the current level to the mean; for an AR(1) process it is exactly $\mu + \phi^h(y_t - \mu)$. We therefore use

$$\text{start}_h = y_{\text{last}} + (\bar{y} - y_{\text{last}})(1 - e^{-h/\tau}), \quad (2)$$

with τ learned (one parameter, initialised at 12 steps). Both required mechanisms are unchanged. The model has $P = 6,626$ parameters.

Training. MSE on the scaled target; AdamW at 3×10^{-4} , halved every epoch; batch 64; gradient clipping at 1.0; at most 10 epochs, early-stopped on validation with patience 3. E counts every epoch trained.

7 Task 2: Experimental Protocol

- **Split.** Chronological 80/10/10 split of the released history. Validation drives early stopping and every decision; test is reported only.
- **Metric.** Forecast origins every 24 steps give 175 overlapping 168-step blocks per split. The headline metric is the mean per-block RMSE, i.e. the expected value of one leaderboard score. MAE and sMAPE are also reported.
- **Seeds.** 5 seeds for the required ablation and the final design; 3 for other variants. Comparisons are paired by seed. 120 runs in total.
- **Leakage tests.** Unit tests verify training-only scaling, that the *past* regime hides future covariates, and that Auto-Correlation matches a direct computation.

8 Task 2: Results

8.1 What the Optional File did

Table 2: External-data ablation: 5 seeds, mean $\pm$ std of block RMSE. Configuration: $d_{\text{model}} = 32$, reference initialisation.

Covariates	Val RMSE	Val MAE	Val sMAPE	Test RMSE
future	70.4 ± 1.6	56.9	64.2	61.6 ± 1.5
past only	91.7 ± 2.3	76.7	72.8	85.8 ± 2.1
none	89.1 ± 1.6	74.4	71.0	88.4 ± 2.3
train mean (reference)	85.9	70.1	71.2	81.6
persistence (reference)	105.6	87.8	80.2	104.5

Future covariates are decisive, here they are worth 18.8 validation RMSE, worse without them on 5/5 seeds, and 26.8 on test (Table 2). Past-only covariates give nothing. The covariates themselves have short memory, so their origin value says little about the coming week. Both history-only variants are worse than the constant mean. Persistence does beat the mean for the first ≈ 12 steps, so this is a limitation of the model rather than of the data (Section 8.3).

8.2 Design Study

Table 3: Paired validation Δ block RMSE vs the $d_{\text{model}}=32$ base (70.4).

Change	Δ	Seeds	Verdict
log(1 + y) target	+8.6	3	worse on every seed
input length 96	+2.8	3	worse on every seed
input length 336	-0.7	3	within noise
covariates as marks	-1.2	3	within noise; test +2.2
kernel 13 / 49	-0.1 / +1.0	3	within noise
$c = 3$ in $k = c \ln L$	+0.1	3	within noise
no phase features	+1.4	3	within noise
$d_{\text{model}} 16 / 64$	+1.2 / +0.8	5 / 3	within noise

Log target. Log-space MSE estimates something close to a conditional median, which sits below the mean for right-skewed data. RMSE therefore worsens by 8.6 while sMAPE improves (53.9 vs 64.2); the metrics disagree as theory predicts.

Other settings. Longer inputs help until memory runs out ($96 < 168 \approx 336$). The Auto-Correlation settings (k , kernel) are immaterial on this weakly periodic series. Model sizes 16–64 are indistinguishable, so we choose $d_{\text{model}} = 16$.

Early design. A first, reference-style design (log target, marks, learning rate 10^{-3}) overfitted within 1–2 epochs and was unstable across seeds (logs in `results/exploratory/`).

8.3 Horizon diagnosis and decoder initialisation

The failure. With lag-1 autocorrelation 0.97, persistence has a step-1 RMSE of 19.4. The reference-initialised model had about 88, and 75 even on training windows. It starts the horizon at the window mean, and its moving averages blur the label/horizon boundary.

Fixed alternatives fail (Table 4).

- Anchoring alone leaves the start at the mean.
- Starting at the last value fixes the first steps but loses the reversion to the mean: block RMSE 89.3.

Table 4: Validation block RMSE and pooled RMSE at selected horizon steps (5 seeds unless noted). Persistence: 19.4 / 57.8 / 107.6 / 136.0 at steps 1 / 6 / 24 / 168.

Configuration	P	Val RMSE	Test RMSE	step 1	step 6	step 24	step 168
d_{model} 32, mean init (reference)	23,489	70.4 ± 1.6	61.6 ± 1.5	≈ 88	≈ 80	≈ 87	≈ 104
d_{model} 32, last-value anchoring	23,681	72.0 ± 2.0	61.5 ± 1.4	85.6	76.1	86.7	99.6
d_{model} 32, start at last value (3 seeds)	23,489	89.3 ± 1.0	84.3 ± 1.5	48.5	65.9	99.1	129.9
d_{model} 32, decay init, Eq. (2)	23,490	69.3 ± 1.5	60.3 ± 0.8	54.3	58.1	80.4	99.0
d_{model} 16, mean init	6,625	71.6 ± 1.1	62.1 ± 1.4	85.9	75.2	87.3	103.0
d_{model} 16, decay init (submitted)	6,626	70.7 ± 1.1	62.9 ± 1.8	47.7	56.7	84.3	103.5

The Decay Initialisation, Eq. (2).

- It improves validation RMSE on 5/5 seeds at both sizes: -1.10 ± 0.31 at $d_{\text{model}} = 32$ and -0.86 ± 0.41 at 16.
- It roughly halves the step-1 error (Fig. 5).
- The learned $\tau = 12.2$ steps matches the target’s autocorrelation half-life.
- On test it helps at $d_{\text{model}} = 32$ (4/5 seeds) and is mixed at 16 ($+0.75 \pm 1.37$). Selection follows validation.

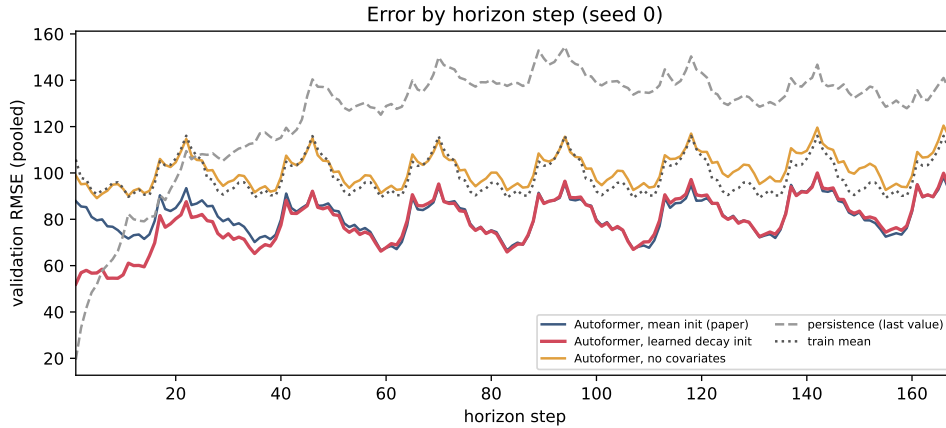


Figure 5: Validation RMSE by horizon step (seed 0). The decay initialisation follows persistence early and the mean-initialised model late.

8.4 Final Model and Leaderboard

The final model trains on time_idx 1–41,472 and early-stops on the last 13 weeks: RMSE 77.2, block std 27.4 (Fig. 6). It uses seed 0, fixed a priori. $P = 6,626$; $E = 6$ (best epoch 3 plus 3 patience epochs; no refit). On the hidden week, RMSE is 101.10, MAE 61.40 and sMAPE 53.79%. MAE is near the backtest (56.8), and sMAPE is better than it (65.2%). And RMSE/MAE is 1.65, against about 1.25 in validation, so a few large misses, most likely unforecast pollution-type peaks in a late-December week, dominate the squared error. One block is a single draw: 101 lies within one standard deviation of the winter-window backtest. The cost penalty adds only ≈ 0.003 to the score.

A 78-parameter ridge on the same inputs (last 24 values, future covariates, phase features) reaches validation 62.9 and test 55.3, better than every Autoformer. It is reported for comparison only, since the rules require an Autoformer. The Autoformer passes the ridge’s training loss within two epochs while its validation error rises, consistent with [3].

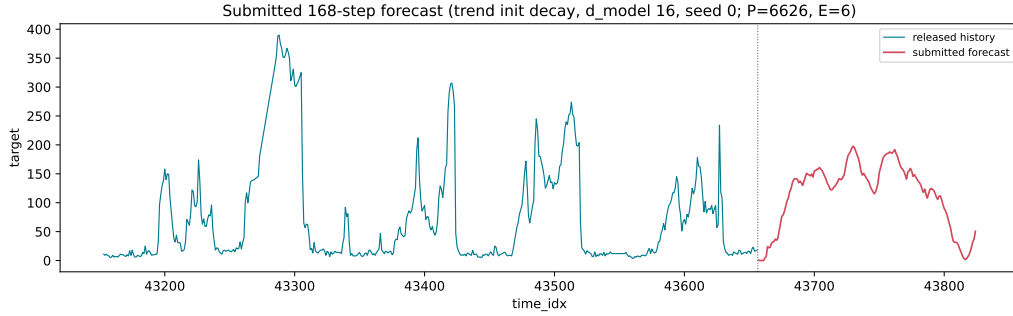


Figure 6: Last three weeks of released history and the submitted 168-step forecast.

9 Discussion

Both tasks give the same answer to the central question. In Task 1, supplying the pace to a linear map beat every learned mixer. In Task 2, a linear model with the right inputs beat Autoformers with 6–88k parameters. The largest Autoformer gain came from a one-parameter structural prior (Eq. (2)), not from added capacity. Most results also agree with theory. We confirmed the shared-map compromise, linear amplitude equivariance and the moving-average leakage. Recurrence appeared at P and $2P$, the two switches were sub-additive, and a log loss produced median-like forecasts. Past-only covariates were useless, as expected under short memory. Three results deviated, and each has an explanation. Attention’s excess held-out loss reflects overfitting to three stations. Prediction 4(a) failed because large-range drift is near-linear. The reference Autoformer’s step-1 failure comes from its decoder initialisation.

10 Conclusion and Limitations.

On a synthetic benchmark with known structure, a period-routed linear model outperformed Attention and Autoformer-style models. It did so on both seen and unseen stations, at a fraction of the cost. On a real, heavy-tailed series, future-known external variables were the dominant source of skill. Correcting the Autoformer’s decoder initialisation with a theory-derived, one-parameter prior improved it consistently. In both settings, choosing the right assumptions mattered more than adding flexibility. As for the limitations these findings have four limitations. Task 1’s neural results use a single data seed. The decomposition width was chosen with an oracle. Task 2’s final model is a single seed, and one leaderboard week is a noisy estimate of its accuracy. Finally, the decay initialisation’s benefit on test at $d_{\text{model}} = 16$ is not conclusive.

References

- [1] H. Wu, J. Xu, J. Wang, and M. Long, “Autoformer: Decomposition transformers with auto-correlation for long-term series forecasting,” in *Proc. NeurIPS*, 2021.
- [2] A. Vaswani *et al.*, “Attention is all you need,” in *Proc. NeurIPS*, 2017.
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11 AI Policy Compliance

Please note that claude and gemini was used for help in coding, formatting i.e. making tables and converting solutions into latex format and understanding of the underlying concepts.

- **Prompt:** Can you explain me the setup of the problem one in simple and intuitive sense.
- **AI Output and corrections:** Explained the complete setup of the problem with the diagram and example.
- **Prompt:** can you explain the complete architecture of the autoformer, how is it different from the transformer, what are the blocks involved in it and explain each block as well as the maths
- **AI Output and corrections:** Explained the complete architecture of autoformer series decomposition, embedding, autocorrelation, encoder layer output and loss function. Also explained the maths behind the autocorrelation block and how it replaces the traditional attention block.
- **Prompt:** Check the results the Raw Attention degrade more on Station 4 than amplitude alone why is that?
- **AI Output and corrections:** Explained that Anchored linear maps scale exactly with amplitude (shared ridge went from 0.731 to 0.876, predicted 0.898). Scaling alone would put Attention near 0.45, but it was 0.621, so it overfitted to three stations.
- **Prompt:** we want to avoid leakage as per my understanding then we are using future covariates as well, is it not like leaking in the future info which we actually wanted to avoid?
- **AI Output and corrections:** Clarified that they're external measurements given on purpose (§2.3), not the target. And ablation shows the future values were worth about 19 RMSE, while past-only values gave nothing because the covariates have short memory.
- **Prompt:** we are doing series decomposition inside the autoformer to separate the seasonality and the trends, when we give data it separate both at the start, why is this block attached at each step, would not the one be enough?
- **AI Output and corrections:** Residual connections and nonlinearities can reintroduce slow structure, so repeated decomposition keeps the seasonal stream clean. It can hurt when the level itself is the signal. That's exactly what your step-1 diagnosis found.